

Polynomial Regression

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Outline

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Polynomial Regression

Learning outcomes

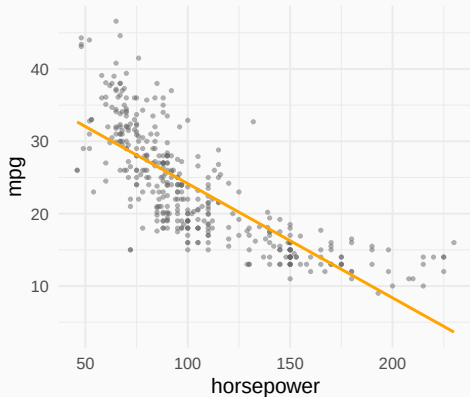
- Recognize a non-linear relationship from a scatter plot and from a residual plot
- Fit polynomial regression as an ordinary linear regression that uses the powers of x as columns of the design matrix
- Interpret a polynomial fit through the slope of the fitted curve and its turning point
- Choose the degree with nested-model F -tests and judge the risks: collinearity of powers, extrapolation, overfitting
- Use other transformations of the features ($\log x$, $\sqrt{x}$) within the same framework

When a straight line is not enough

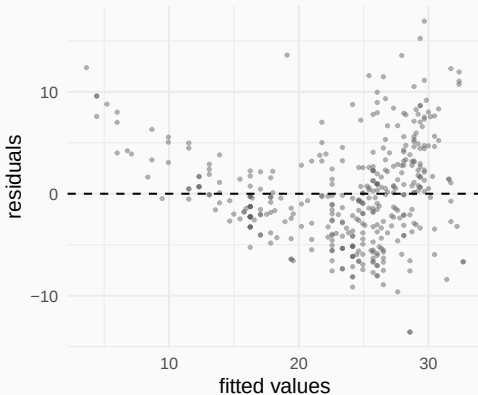
- Auto data: fuel economy (mpg) against engine power (horsepower) for $n = 392$ cars
- The scatter plot is curved: mpg falls quickly at low horsepower and levels off at high horsepower
- The straight-line fit has $R^2 = 0.606$, but its residuals show a U-shape: they are positive at both ends of the fitted range and negative in the middle, so the line under-predicts at the extremes and over-predicts in between
- A systematic pattern in the residuals means the model has missed structure that the data contain

When a straight line is not enough (cont.)

straight-line fit



residuals vs fitted values



The polynomial regression model

- **Polynomial regression** adds powers of the feature:
 $y = \beta_0 + \beta_1 x + \beta_2 x^2 + \dots + \beta_d x^d + \varepsilon$; d is the *degree*
- It is still a *linear* regression: the model is linear in the coefficients $\beta_0, \dots, \beta_d$; the columns $x, x^2, \dots, x^d$ are simply treated as d separate features
- Any continuous curve on a bounded interval can be approximated arbitrarily well by a polynomial of high enough degree (Weierstrass theorem), so polynomials are a flexible family
- Degree 1 is a line, degree 2 a parabola (one bend), degree 3 an S-shape (up to two bends); a degree- d polynomial has at most $d - 1$ bends

Ordinary least squares still applies

- Only the columns of the design matrix change; the first cars in the Auto data (horsepower 130, 165, 150 and mpg 18, 15, 18) give, for $d = 2$:

$$\tilde{\mathbf{X}} = \begin{bmatrix} 1 & x^{(1)} & (x^{(1)})^2 \\ 1 & x^{(2)} & (x^{(2)})^2 \\ 1 & x^{(3)} & (x^{(3)})^2 \\ \vdots & \vdots & \vdots \end{bmatrix} = \begin{bmatrix} 1 & 130 & 16900 \\ 1 & 165 & 27225 \\ 1 & 150 & 22500 \\ \vdots & \vdots & \vdots \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 18 \\ 15 \\ 18 \\ \vdots \end{bmatrix}$$

- The normal equations, $\hat{\beta} = (\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} \tilde{\mathbf{X}}^\top \mathbf{y}$, and everything derived from them carry over: standard errors, t -tests, the F -test, ANOVA, confidence and prediction intervals, residual diagnostics
- The powers of x are functions of the same variable, so features cannot be changed one at a time: "holding the other features fixed" has no meaning for x and x^2

The quadratic fit

	coefficient	std. error	<i>t</i> - statistic	<i>p</i> - value
intercept	56.9001	1.8004	31.60	< 0.0001
horsepower	-0.46619	0.03112	-14.98	< 0.0001
horsepower ²	0.001231	0.000122	10.08	< 0.0001

- $\widehat{\text{mpg}} = 56.90 - 0.4662 \text{ hp} + 0.001231 \text{ hp}^2$, with $R^2 = 0.688$ (linear fit: 0.606) and $\text{RSE} = 4.37$ (linear fit: 4.91)
- The overall F -test gives $F = 428.0$ on $(2, 389)$ degrees of freedom, $p < 0.0001$
- The quadratic term is highly significant ($t = 10.08$): the curvature is real, not noise

Interpreting a polynomial

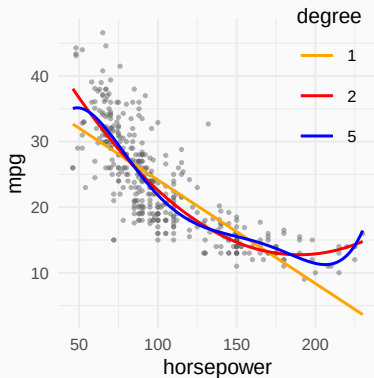
- The effect of the feature depends on its own value: for the quadratic fit the slope is $\partial\hat{y}/\partial x = \hat{\beta}_1 + 2\hat{\beta}_2x$, like an interaction of x with itself
- Auto: the slope is -0.343 mpg per horsepower at 50 hp, -0.220 at 100 hp, -0.097 at 150 hp and $+0.100$ at 230 hp: extra power costs less and less mileage, and eventually the fit turns upward
- $\hat{\beta}_2 > 0$ means the curve is convex (U-shaped), $\hat{\beta}_2 < 0$ concave (hill-shaped); individual coefficients such as $\hat{\beta}_1 = -0.466$ are not the effect of a one-unit change in x
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Polynomial fits on the Auto data

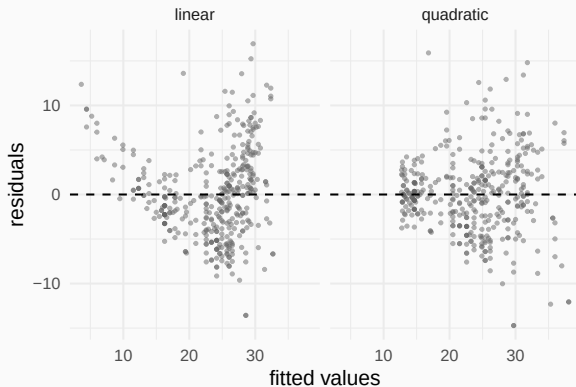
- R^2 is 0.606 for the linear fit, 0.688 for the quadratic and 0.697 for degree 5: the quadratic term helps a lot, further terms barely
- The linear fit leaves a U-shaped pattern in the residuals, a systematic structure it failed to capture; for the quadratic fit the U-shape is gone
- Higher degrees always improve the fit to the training data, but degree 5 starts to wiggle at the ends: too much flexibility is a risk (bias - variance trade - off, later)

Polynomial fits on the Auto data (cont.)

fits of degree 1, 2 and 5



residuals vs fitted values



Choosing the degree with nested F-tests

degree d	RSS	R^2	adjusted R^2	F vs degree $d - 1$	p -value
1	9385.9	0.6059	0.6049		
2	7442.0	0.6876	0.6860	101.6	< 0.0001
3	7426.4	0.6882	0.6858	0.82	0.367
4	7399.5	0.6893	0.6861	1.41	0.236
5	7223.4	0.6967	0.6928	9.41	0.0023

- Each row tests $H_0: \beta_d = 0$ with the partial F -test
 $F = (\text{RSS}_{d-1} - \text{RSS}_d) / (\text{RSS}_d / (n - d - 1))$, which equals the squared t -statistic of the degree- d term
- Going from degree 1 to 2 removes 21% of the RSS; degrees 3 and 4 add nothing significant; the degree-5 term has $p = 0.0023$

Which degree? Significance is not usefulness

- A stopping rule “add terms until the first non-significant one” picks degree 2; but the degree-5 term is significant after two non-significant ones, and adjusted R^2 is highest at degree 5 (0.6928)
- Yet degree 5 lowers the RSS by only 2.4% and raises R^2 by only 0.007, and its curve wiggles at the ends of the data
- With $n = 392$ even small improvements can be significant, and several tests are run on the same data
- In-sample criteria alone cannot settle the degree; combine them with the fitted curve and residual plots, and evaluate on new data (later lecture)

Collinearity among the powers

- x and x^2 move together: in the Auto data $\text{cor}(\text{hp}, \text{hp}^2) = 0.983$ and both have $\text{VIF} = 29.3$, well above the warning level of 5 to 10
- **Centering** the feature removes most of this: with $\tilde{x} = \text{hp} - 104.47$ (the mean), $\text{cor}(\tilde{x}, \tilde{x}^2) = 0.663$ and $\text{VIF} = 1.78$
- The fit itself does not change (same R^2 , RSE and fitted values, same $\hat{\beta}_2 = 0.001231$); the intercept becomes the predicted mpg at the mean horsepower (21.63) and $\hat{\beta}_1 = -0.2091$ (SE 0.0077) becomes the slope at the mean horsepower
- **Orthogonal polynomials**: go one step further and make the columns exactly uncorrelated; the fitted curve is the same, but the individual coefficients are no longer interpretable
- Raw powers also cause numerical problems (the values of x^5 reach 6×10^{11} here), another reason to center or use orthogonal polynomials

Other transformations of the features

model for mpg	parameters	R^2
hp	2	0.606
$\log(\text{hp})$	2	0.668
$\sqrt{\text{hp}}$	2	0.644
$1/\text{hp}$	2	0.667
$\text{hp} + \text{hp}^2$	3	0.688

- Any fixed function of a feature can be a column of $\tilde{\mathbf{X}}$ ($\log x$, $\sqrt{x}$, $1/x$, x^2 , ...): the model stays linear in the coefficients
- A one-parameter transformation such as $\log(\text{hp})$ captures much of the curvature with one coefficient fewer than a quadratic
- Choose among candidates with residual plots and subject-matter knowledge, not R^2 alone

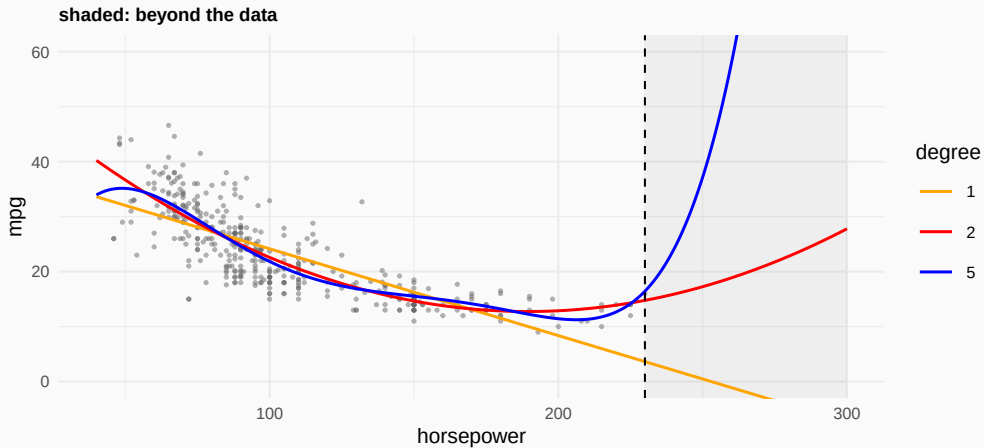
Extrapolation and overfitting

- A polynomial is a global curve, and outside the range of the data it can behave wildly: the Auto data end at 230 hp, yet the quadratic fit predicts 27.8 mpg at 300 hp (more than at 100 hp: 22.6) and the degree-5 fit predicts 278 mpg
- Each added power makes the curve more flexible and lowers the training error, whether or not the new term reflects reality: this is **overfitting**, the fit follows noise
- Degree 5 already turns twice inside the data range and rises at the right end; ISLP notes that polynomial degrees above 3 or 4 are rarely used

Extrapolation and overfitting (cont.)

- Deciding how much flexibility the data support requires new data or resampling: bias - variance trade - off, validation and cross - validation (later lectures)

Extrapolation and overfitting (cont.)



Summary

Summary

- **Non-linearity** shows up as a pattern in the residuals; polynomial regression $y = \beta_0 + \beta_1 x + \dots + \beta_d x^d + \varepsilon$ is a linear regression on the design matrix with columns $1, x, \dots, x^d$
- **Interpretation:** the effect of x depends on x itself (slope $\hat{\beta}_1 + 2\hat{\beta}_2 x$ for $d = 2$)
- **Degree:** nested F -tests, adjusted R^2 and residual plots guide the choice, but they are in-sample tools
- **Cautions:** collinearity of powers (center or use orthogonal polynomials), extrapolation, overfitting
- **Other transformations:** $\log x$, $\sqrt{x}$, $1/x$ fit the same framework

Questions?
